# QR Code-Based Secure Authentication System

**A CAPSTONE**

**PROJECT REPORT**

*Submitted in the partial fulfilment for the Course of*

**CSA5115-Crytptography and Network Security**

*to the award of the degree of*

**BACHELOR OF ENGINEERING**

**IN**

**COMPUTER SCIENCE ENGINEERING**

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# SIMATS ENGINEERING


**Saveetha Institute of Medical and Technical Sciences**

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# DECLARATION

We hereby declare that the report entitled **" QR Code-Based Secure Authentication System"** is a unique and original work submitted by us for the degree of **Bachelor of Engineering**. This report is the record of our capstone project for the course **CSA5115-Crytptography and Network Security**, carried out under the guidance of **Dr.T. Sangeetha.** We further declare that this work has not formed the basis for the award of any degree or diploma in this or any other University or institution of higher learning.

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# BONAFIDE CERTIFICATE

Certified that this capstone project reports on **"QR Code-Based Secure Authentication System"** is the Bonafide work Sweta.R (192511408), Elakati Renu Sree(192511035),Tarunika.S(192511024), Abinaya.R(192421161)who carried out the capstone project work under my supervision for the course **CSA5115-Crytptography and Network Security.**

**SIGNATURE**

**Dr. T. Sangeetha**

**SUPERVISOR**

Submitted for the Project work Viva-Voce held on ______________________.

INTERNAL EXAMINER　　　　　　　　　　EXTERNAL EXAMINER

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Sincerely,

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# ABSTRACT

The **QR Code-Based Secure Authentication System** is a modern authentication solution designed to verify a user's identity through QR codes in a secure, fast, and convenient manner. Traditional login systems generally require users to enter passwords repeatedly, which can be inconvenient and may expose authentication credentials to security threats. The proposed system provides an alternative approach by generating a unique QR code for each login request, allowing users to authenticate by scanning the code using a mobile device.

The primary objective of this project is to develop a secure authentication mechanism that combines QR code technology with encryption and verification techniques. The system consists of four major modules: **User Registration, QR Code Generation, Encrypted QR Verification, and Login Authentication & Activity Log**. During registration, the user provides personal details such as name, email, phone number, and password. The system validates the information, securely stores the password, and creates a unique user account.

For every login request, the system generates a unique QR code containing an encrypted authentication token. The QR code is displayed on the login screen and remains valid only for a limited period, reducing the possibility of unauthorized reuse. When the user scans the QR code using a mobile device, the system decrypts the QR data and verifies its authenticity. It also checks whether the QR code has expired or has already been used before granting access.

The system additionally maintains an activity log containing login and logout information, including the date, time, and authentication status. The project uses **Python 3.x, Flask, SQLite3 or MySQL, qrcode, pyzbar, OpenCV, cryptography (Fernet), bcrypt, and Visual Studio Code** for development.

The proposed system improves authentication security, reduces repeated password entry, and provides a user-friendly login experience. It can be applied in areas such as banking, online services, educational institutions, and office access systems, where secure and convenient user authentication is required.

# CHAPTER 1

# INTRODUCTION

## 1.1 Background Information

With the rapid growth of online services and digital applications, secure user authentication has become increasingly important for protecting personal information and preventing unauthorized access. Traditional authentication methods mainly rely on usernames and passwords, which may be difficult to manage and can create security risks when credentials are exposed. QR-based authentication systems provide a convenient alternative by allowing users to verify their identity through QR codes. The system generates a unique QR code for each login request and uses encrypted authentication data to improve security. This project focuses on developing a QR Code-Based Secure Authentication System that provides fast, secure, and user-friendly authentication through QR code generation, verification, and activity logging.

## 1.2 Project Objectives

The main objective of this project is to design and develop a secure authentication system using QR codes. The project aims to provide quick and reliable user verification without requiring repeated password entry. Another goal is to generate unique and encrypted QR codes for each login request and verify their authenticity before granting access. The system also aims to check QR code validity, prevent reuse of expired codes, securely manage user information, and maintain login activity records. Overall, the project focuses on improving authentication security, reducing unauthorized access, and providing a simple and user-friendly login experience.

## 1.3 Significance of the Project

This project is significant because it demonstrates how QR code technology and encryption techniques can be applied to improve secure user authentication. By generating unique QR codes for login requests, the system can reduce the risks associated with repeated password entry and unauthorized access. The project provides a practical approach to secure authentication by combining QR verification, encrypted authentication tokens, and activity logging. It also benefits users by providing a faster and more convenient login experience while maintaining security. The system can be useful in banking, online services, educational institutions, and office access systems.

## 1.4 Scope of the Project

The scope of this project includes the development of a QR Code-Based Secure Authentication System that enables users to register, generate QR codes, verify encrypted authentication data, and securely access the system. The project focuses on generating unique and time-limited QR codes for login requests and maintaining activity logs for successful and failed authentication attempts. However, it does not include advanced biometric authentication, large-scale cloud deployment, or integration with external authentication services. The project is limited to demonstrating secure QR-based authentication using the proposed modules and technologies.

## 1.5 Methodology Overview

The project follows a structured methodology consisting of user registration, QR code generation, encrypted QR verification, and login authentication with activity logging. Initially, the user provides personal details and the system validates the entered information before creating an account. During login, a unique QR code containing an encrypted authentication token is generated and displayed. The user scans the QR code using a mobile device, after which the system decrypts and verifies the authentication data. The system then checks the QR code validity and usage status before granting access. Finally, login and logout activities are recorded for security monitoring and future reference.

# CHAPTER 2

# PROBLEM IDENTIFICATION AND ANALYSIS

## 2.1 Description of the Problem

The increasing use of online services has created a greater need for secure and convenient user authentication. Traditional password-based authentication can be inconvenient and may expose users to risks when passwords are weak, reused, or compromised. Users also need to repeatedly enter their credentials during login, which can reduce convenience and increase the possibility of unauthorized access. This project addresses these challenges by developing a QR Code-Based Secure Authentication System that uses unique QR codes and encrypted authentication tokens to verify users. The system provides a secure method for authentication while reducing dependence on repeated password entry and improving the overall login experience.

## 2.2 Evidence of the Problem

The need for secure authentication is increasing as more services require users to access digital platforms using personal accounts. Password-based systems may create security concerns when credentials are exposed or repeatedly entered during login. The proposed QR-based approach addresses these concerns by generating a unique QR code for each login request and using an encrypted authentication token for verification. The system also checks whether the QR code has expired or has already been used, helping prevent unauthorized reuse. These features provide evidence of the need for an authentication method that is secure, convenient, and capable of monitoring login activities.

## 2.3 Stakeholders

Several groups are involved in the QR Code-Based Secure Authentication System. The primary stakeholders are users who require secure and convenient access to online services. System administrators are also important stakeholders, as they manage user accounts, authentication activities, and security monitoring. Developers and security professionals are responsible for designing, implementing, and maintaining the authentication system. Organizations such as banks, educational institutions, online services, and offices can benefit from secure QR-based authentication for protecting their systems and user information.

## 2.4 Supporting Data and Research

Research on authentication systems shows that QR codes can provide a convenient method for user verification by transferring authentication information between devices. The proposed system uses encrypted authentication tokens within QR codes to improve security during the login process. The system also verifies whether the QR code is authentic, valid, and unused before granting access. Technologies such as QR code generation, QR code scanning, encryption, password security, and activity logging support the development of the proposed authentication system. These approaches provide a practical foundation for developing a secure and user-friendly QR-based authentication solution.

# CHAPTER 3

# SOLUTION DESIGN AND IMPLEMENTATION

### 3.1 Development and Design Process

The development of the QR Code-Based Secure Authentication System followed a structured process. Initially, the authentication requirements were analyzed and the system modules were identified. The user registration module was designed to collect and validate personal information and securely store passwords. The QR code generation module was then developed to create a unique QR code containing an encrypted authentication token for each login request. The verification module was designed to decrypt and validate the QR code and check its expiry and usage status. Finally, the login authentication and activity log module was implemented to grant access only after successful verification and record authentication activities. The complete system was tested to ensure secure and reliable authentication.

### 3.2 Tools and Technologies Used

The following tools and technologies were utilized in the development of the project:

- Programming Language: Python 3.x
- Framework: Flask
- Database: SQLite3 or MySQL
- QR Code Generation: qrcode
- QR Code Scanning: pyzbar
- Image Processing: OpenCV (cv2)
- Encryption: cryptography (Fernet)
- Password Security: bcrypt
- Development Environment: Visual Studio Code

These technologies provide the required functionality for user registration, QR code generation and scanning, encrypted verification, authentication, and activity logging.

### 3.3 Solution Overview

The proposed system is a QR-based authentication platform designed to verify users securely through unique QR codes. The system first registers the user and validates the provided information. During login, a unique QR code containing an encrypted authentication token is generated and displayed on the login screen. The user scans the QR code using a mobile device, and the system decrypts and verifies the authentication data. The system also checks whether the QR code has expired or has already been used before granting access.

The system architecture consists of four main components:

1. User Registration Module – Collects and validates user information and creates an account.
2. QR Code Generation Module – Generates a unique, encrypted QR code for each login request.
3. Encrypted QR Verification Module – Decrypts and verifies the QR code and checks its validity.
4. Login Authentication & Activity Log Module – Grants access after successful authentication and records login activities.

This design provides secure authentication while maintaining convenience and supporting security monitoring.

### 3.4 Engineering Standards Applied

The project follows recognized software and security practices to maintain reliability, security, and consistency throughout development. Secure password handling is applied using bcrypt, while encrypted authentication data is protected using the Fernet encryption mechanism. QR codes are generated with unique authentication tokens and are designed to remain valid for a limited period. The system also verifies whether a QR code has expired or has already been used before allowing authentication. Proper validation and activity logging are applied to improve system reliability and support security monitoring. These practices help ensure that the authentication system provides secure and controlled access to authorized users.

### 3.5 Solution Justification

The proposed QR Code-Based Secure Authentication System provides an effective solution to the limitations of traditional password-based authentication. Unique QR codes reduce the need for repeated password entry, while encrypted authentication tokens help protect the information used during login. The limited validity of each QR code and verification of previously used codes provide additional protection against unauthorized reuse. The activity log also helps monitor successful and failed authentication attempts. By combining QR code technology, encryption, password security, and authentication monitoring, the system provides a secure, convenient, and user-friendly approach to identity verification.

# CHAPTER 4

# RESULTS AND RECOMMENDATIONS

## 4.1 Evaluation of Results

The QR Code-Based Secure Authentication System was successfully designed to provide secure and convenient user authentication through QR codes. The system allows users to register their details, generate unique QR codes for login requests, and verify encrypted authentication data before granting access. The QR codes are checked for validity and previous usage, helping prevent unauthorized reuse. The activity log also records successful and failed login attempts for security monitoring. The results indicate that the proposed system can provide faster authentication, reduce repeated password entry, and improve the overall security and convenience of the login process.

## 4.2 Challenges Encountered

Several challenges were encountered during the development of the authentication system. One major challenge was securely generating and handling unique QR codes for each login request. Another difficulty was implementing encryption and decryption while ensuring that the authentication data could be verified correctly. Managing QR code expiry and preventing previously used codes from being accepted also required careful implementation. Integrating QR code scanning with the authentication process and maintaining accurate login activity records were additional challenges. These issues were addressed through proper validation, encryption techniques, and systematic testing of each authentication module.

## 4.3 Possible Improvements

Although the system achieves its main authentication objectives, certain improvements can be introduced in the future. The system can be enhanced by integrating biometric authentication such as fingerprint or facial verification along with QR authentication. Cloud-based database support can also be introduced to improve scalability and accessibility. Additional security mechanisms such as multi-factor authentication and stronger token management can further improve protection against unauthorized access. A dedicated mobile application can also be developed to provide faster QR scanning and authentication. These improvements can make the system more secure, scalable, and suitable for larger real-world applications.

### 4.4 Recommendations

For future development, it is recommended to use strong encryption and secure password-storage techniques to protect user information. QR codes should always be generated uniquely and configured with limited validity to prevent unauthorized reuse. Organizations implementing the system should maintain proper authentication logs and regularly monitor failed login attempts for suspicious activity. Multi-factor authentication can also be incorporated to provide an additional layer of security. Regular security testing and software updates should be performed to identify vulnerabilities and maintain system reliability. These practices can help ensure secure, efficient, and user-friendly authentication.

# CHAPTER 5

# REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

## 5.1 Key Learning Outcomes

The development of the QR Code-Based Secure Authentication System provided an opportunity to apply various concepts learned throughout the academic curriculum. The project strengthened my understanding of authentication systems, QR code technology, encryption, password security, and secure user verification. It also enhanced my knowledge of system design, user registration, QR code generation, and authentication activity logging. By implementing these concepts in a practical application, I gained a deeper understanding of how security techniques can be combined to solve real-world authentication problems efficiently.

## 5.2 Challenges Encountered and Overcome

### Personal and Professional Growth

One of the major challenges during the project was designing a secure authentication process that could generate and verify unique QR codes correctly. Another challenge was handling encrypted authentication tokens and checking whether QR codes were expired or already used. At times, testing the registration, scanning, verification, and login processes required careful debugging. Overcoming these challenges improved my confidence, patience, and problem-solving abilities. The experience also taught me the importance of testing each module carefully and maintaining security throughout the development process.

## 5.3 Application of Engineering Standards

The project followed appropriate software development and security practices to ensure the reliability and effectiveness of the authentication system. Proper validation, secure password handling, encrypted authentication data, and activity logging were considered during the development process. The use of technologies such as Flask, cryptography, bcrypt, and database systems supported the implementation of secure authentication features. Following structured development and security practices helped maintain consistency, improve system reliability, and reduce potential authentication risks during the implementation of the project.

## 5.4 Insights into the Industry

Working on this project provided valuable insights into the importance of secure authentication in modern digital services. I learned how authentication systems can combine QR codes, encryption, password security, and activity monitoring to protect user accounts. The project also highlighted the importance of convenience and security when designing systems for real-world users. QR-based authentication can be useful in areas such as banking, educational institutions, online services, and office access systems. This experience increased my awareness of industry requirements and encouraged me to explore advanced cybersecurity technologies.

## 5.5 Conclusion of Personal Development

The capstone project significantly contributed to my academic, technical, and personal development. It strengthened my understanding of secure authentication, QR code technology, encryption, and software development while improving my programming and problem-solving skills. The experience taught me valuable lessons in system design, testing, debugging, and security-conscious development. It also improved my confidence in applying theoretical concepts to practical problems. Overall, this project has provided useful technical knowledge and practical experience that will support my future learning and professional development in software development and cybersecurity.

# CHAPTER 6

# CONCLUSION

The QR Code-Based Secure Authentication System was developed to address the challenges associated with traditional password-based authentication. The proposed system uses QR codes to provide users with a secure, fast, and convenient method of verifying their identity. By generating a unique QR code for each login request and using encrypted authentication tokens, the system reduces the risk of unauthorized access and repeated use of authentication information.

The implementation of the system consists of four major modules: User Registration, QR Code Generation, Encrypted QR Verification, and Login Authentication & Activity Log. The system validates user details during registration, generates time-limited QR codes during login, verifies the encrypted QR data, and checks whether the QR code has expired or has already been used. The activity logging mechanism also records login and logout information, supporting security monitoring and future reference.

The significance of this project lies in its practical application of QR code technology, encryption, and secure authentication techniques. It provides a user-friendly alternative to conventional password-based login while maintaining an additional level of security. The system can be applied in banking, online services, educational institutions, and office access systems where reliable user authentication is required.

Overall, the project demonstrates that QR-based authentication can provide an efficient and convenient approach to secure login. The proposed system successfully combines authentication, encryption, QR verification, and activity monitoring to create a reliable security mechanism. Future enhancements can further improve the system by integrating additional authentication methods and advanced security features.

## REFERENCES


1. Temoshok, D., Choong, Y.-Y., Regenscheid, A., Galluzzo, R., Fenton, J. L., Richer, J., & Lefkovitz, N. (2025). *NIST SP 800-63B-4: Digital Identity Guidelines – Authentication and Authenticator Management*. National Institute of Standards and Technology.
2. Temoshok, D., Choong, Y.-Y., Galluzzo, R., LaSalle, M., Regenscheid, A., Proud-Madruga, D., Gupta, S., & Lefkovitz, N. (2025). *NIST SP 800-63-4: Digital Identity Guidelines*. National Institute of Standards and Technology.
3. Richer, J., Fenton, J. L., Lefkovitz, N., Temoshok, D., Galluzzo, R., & Choong, Y.-Y. (2025). *NIST SP 800-63C-4: Digital Identity Guidelines – Federation and Assertions*. National Institute of Standards and Technology.
4. Lodderstedt, T., Bradley, J., Labunets, A., & Fett, D. (2025). *RFC 9700: Best Current Practice for OAuth 2.0 Security*. Internet Engineering Task Force.
5. OWASP Foundation. (2025). *Authentication Cheat Sheet*. OWASP Cheat Sheet Series.
6. OWASP Foundation. (2025). *Credential Stuffing Prevention Cheat Sheet*. OWASP Cheat Sheet Series.
7. National Institute of Standards and Technology. (2025). *NIST SP 800-63A-4: Digital Identity Guidelines – Identity Proofing and Enrollment*. NIST.
8. National Institute of Standards and Technology. (2025). *NIST Digital Identity Guidelines, Revision 4*. NIST.
9. Python Software Foundation. (2025). *Python Documentation*. Python Software Foundation.
10. Pallets Projects. (2025). *Flask Documentation*. Pallets Projects.
11. OpenCV Team. (2025). *OpenCV Documentation*. Open Source Computer Vision Library.
12. PyCA Cryptography Developers. (2025). *Cryptography Documentation*. Python Cryptographic Authority.
13. QR Code Developers. (2025). *QR Code Generation and Encoding Documentation*.


**Engineering Standards Referenced**

- **NIST SP 800-63B-4** – Digital Identity Guidelines: Authentication and Authenticator Management
- **NIST SP 800-63A-4** – Digital Identity Guidelines: Identity Proofing and Enrollment
- **NIST SP 800-63C-4** – Digital Identity Guidelines: Federation

## Appendix A: Code Snippets

```
# QR Code Generation

import qrcode
from cryptography.fernet import Fernet

# Generate encryption key
key = Fernet.generate_key()
cipher = Fernet(key)

# Authentication token
token = "USER_LOGIN_AUTHENTICATION_TOKEN"

# Encrypt authentication token
encrypted_token = cipher.encrypt(token.encode())

# Generate QR Code
qr = qrcode.QRCode(
    version=1,
    box_size=10,
    border=5
)

qr.add_data(encrypted_token.decode())
qr.make(fit=True)

# Create and save QR Code
qr_image = qr.make_image()
qr_image.save("authentication_qr.png")

print("Secure QR Code generated successfully.")
```